

- Chapter 1 Complex numbers
2 analytic functions
3. Cauchy integral < chap 3
chap 4

Chapter 5 Infinite Series

$$x_1, x_2, x_3, \dots, x_n, \dots$$

$$1, \frac{1}{2}, \left(\frac{1}{2}\right)^2, \left(\frac{1}{2}\right)^3, \dots$$

$$\sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n$$

$$S_N = \sum_{n=0}^N \left(\frac{1}{2}\right)^n$$

$$N \rightarrow \infty \quad \lim_{N \rightarrow \infty} S_N = C \quad \text{finite}$$

infinite series converges 收敛 convergence
diverge 发散 divergence

$$\sum_{n=0}^N a^n = \frac{1-a^{N+1}}{1-a}$$

$$\sum_{n=n_1}^{n_2} a^n = \frac{a^{n_1} - a^{n_2+1}}{1-a}$$

$$a = \frac{1}{2}$$

$$\sum_{n=0}^{\infty} \frac{1}{2}^n = \frac{1-0}{1-\frac{1}{2}} = 2$$

$$\textcircled{1} \quad \sum_{n=0}^{\infty} u_n \quad \underbrace{u_0 + u_1 + u_2 + \dots + u_n + \dots}_{\uparrow} = C$$

$$\Rightarrow \lim_{n \rightarrow \infty} u_n = 0$$

$$S_N = \sum_{n=0}^N u_n \quad \lim_{N \rightarrow \infty} S_N = C$$

$$T_N = \sum_{n=0}^{N+1} u_n \quad \lim_{N \rightarrow \infty} T_N = C$$

$$T_N - S_N = u_{N+1} \quad \lim_{N \rightarrow \infty} T_N - S_N = 0$$

Test/decide if the infinite series converges

① Comparison test

if $\{b_n\}$ converges

and $|a_n| \leq b_n$,

then $\{a_n\}$ converges

$$\sum_{n=0}^{\infty} b_n = C$$

$$b_n > 0$$

Ex: $\left\{\frac{1}{n!}\right\} \quad n! = 1 \times 2 \times 3 \times \dots \times n$
n factorial

$$\frac{1}{1 \times 2 \times 3 \times 4} < \frac{1}{2 \times 2 \times 2 \times 2}$$

$$\frac{1}{n!} < \frac{1}{2^n}$$

Ex 2 $\left\{\frac{1}{n(n+1)}\right\} \quad n = 2, 3, \dots$

$$\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} \dots$$

$$= \frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \frac{1}{4} - \frac{1}{5} \dots \quad \frac{1}{n} - \frac{1}{n+1}$$

$$= 1 - \frac{1}{n}$$

$$= \frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \frac{1}{4} - \frac{1}{5} \dots - \frac{1}{n} + \frac{1}{n-1}$$

$$= 1 - \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} S_n = 1$$

$$\frac{1}{n!} < \left(\frac{1}{n \cdot (n-1)} \right)$$

② if $\{b_n\}$ diverges, $|a_n| > b_n$
then $\{a_n\}$ must diverge

$$b_n > 0$$

$$\text{Ex 3} \quad \sum_{n=1}^{\infty} \frac{1}{n}$$

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{n} = S \approx \infty$$

a_1	1		1	b_1
a_2	$\frac{1}{2}$		$\frac{1}{2}$	b_2
a_3	$\frac{1}{3} + \frac{1}{4}$	$>$	$\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$	b_3
a_4	$\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}$	$>$	$\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{1}{2}$	b_4
a_5	$\frac{1}{9} + \dots + \frac{1}{16}$	$>$	$\frac{1}{16} + \frac{1}{16} + \dots + \frac{1}{16} = \frac{1}{2}$	b_5

$$\frac{a_1 + a_2 + a_3 + \dots}{\sum a_n} = S \quad 1 + \sum_{n=2}^{\infty} \frac{1}{2} = \infty$$

$$\text{Ex 4} \quad \sum_{n=1}^{\infty} \frac{1}{n^2} < \sum_{n=1}^{\infty} \frac{1}{n(n+1)}$$

converges!

③ Ratio Test

Given $\{u_n\}$. if $\left| \frac{u_{n+1}}{u_n} \right| < \rho < 1$

ρ doesn't depend on n

then, $\{u_n\}$ converges!

$$\text{Ex 5} \quad \left\{ \frac{1}{n!} \right\}$$

$$\left| \frac{u_{n+1}}{u_n} \right| = \frac{1}{n+1} < \frac{1}{2} < 1 \quad n > 1$$

④ Cauchy Test $\sqrt[n]{|u_n|} < \rho < 1$

Given a sequence $\{u_n\}$

if $\lim_{n \rightarrow \infty} |u_n|^{1/n} = c < 1$ then $\{u_n\}$ converges.

$$\text{Ex 6} \quad \sum_{n=1}^{\infty} \frac{i^n}{n}$$

$$\frac{2^1}{n^2}$$

$$\frac{n^n}{n!}$$

$$\frac{n^2}{n^3+1}$$

Z-transform

The infinite series of Z

Given a sequence of numbers

$$\begin{array}{c|cccc} \infty & x_1 & x_2 & x_3 & \dots \\ \hline 0 & 0.5 & 3 & -2 & 1 \dots \end{array} \quad \text{Causal}$$

x_n is causal if $x_n = 0$ for $n < 0$

finite: 0.5 3 -2, 1 0 0 0 0 0

infinite

$$\mathcal{Z}\{x\} = \sum_{n=0}^{\infty} x_n \cdot z^{-n} = X(z)$$

$$x_n: 0.5 \ 3 \ -2, \ 1 \quad \checkmark$$

$$X(z) = 0.5 + 3z^{-1} - 2z^{-2} + 1 \cdot z^{-3}$$

entries

Ex 2 Haar wavelet $\{\frac{1}{2}, \frac{1}{2}\}$ H_0

$\{\frac{1}{2}, -\frac{1}{2}\}$ H_1

$$H_0(z) = \frac{1}{2} + \frac{1}{2}z^{-1}$$

$$H_1(z) = \frac{1}{2} - \frac{1}{2}z$$

Ex $u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$

$$x[n] = a^n u[n]$$

Causal signal

$$x[n] = \left(\frac{1}{2}\right)^n u[n]$$

$$1 \quad \frac{1}{2} \quad \frac{1}{4} \quad \frac{1}{8} \quad \frac{1}{16} \dots$$

$$X(z) = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n \cdot z^{-n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2}z^{-1}\right)^n$$

$$= \frac{1 - \left(\frac{1}{2}z^{-1}\right)^{\infty}}{1 - \frac{1}{2}z^{-1}} \rightarrow 0$$

$$= \frac{1 - 0}{1 - \frac{1}{2}z^{-1}}$$

$$= \frac{z}{z - \frac{1}{2}}$$

$$\sum_{n=n_1}^{n_2} a^n = \frac{a^{n_1} - a^{n_2+1}}{1 - a}$$

$$\left|\frac{1}{2}z^{-1}\right| < 1 \quad \text{condition}$$

$$\left|\frac{1}{2z}\right| < 1$$

$$|z| > \frac{1}{2}$$

$$\left(\frac{1}{2}\right)^n u[n] \longleftrightarrow \frac{z}{z - \frac{1}{2}}, \quad |z| > \frac{1}{2}$$

o $a^n u[n] \longleftrightarrow \frac{z}{z - a} \quad |z| > |a|$

a can be complex

$$\delta[n] = \begin{cases} 1 & n=0 \\ 0 & n \neq 0 \end{cases}$$

$$\delta[n] \longleftrightarrow 1$$

$$u[n] \leftrightarrow \frac{z}{z-1}$$

$$s[n-5] \leftrightarrow z^{-5}$$

$$s[n-6] \leftrightarrow z^{-6}$$

$$a = e^{i\theta}$$

$$e^{i\theta} u[n] \leftrightarrow \frac{z}{z - e^{i\theta}}$$

$$x_1 \ x_2 \ x_3 \ \dots \ x_n \ \dots \leftrightarrow X(z)$$

$$y_1 \ y_2 \ y_3 \ \dots \ y_n \ \dots \leftrightarrow Y(z)$$

$$x_1+y_1 \ x_2+y_2 \ x_3+y_3 \ \dots \ x_n+y_n \leftrightarrow X(z) + Y(z)$$

$$\sum (x_n + y_n) z^{-n} = \sum x_n z^{-n} + \sum y_n z^{-n}$$

$$e^{-i\theta} u[n] \leftrightarrow \frac{z}{z - e^{-i\theta}}$$

$$[e^{i\theta} + e^{-i\theta}] u[n] \leftrightarrow \frac{z}{z - e^{i\theta}} + \frac{z}{z - e^{-i\theta}} \quad \checkmark$$

$$2 \cos \theta u[n] \leftrightarrow$$

$$\sin \theta u[n] \leftrightarrow$$

$$\checkmark \left(\frac{1}{2}\right)^n u[n] + \left(\frac{1}{3}\right)^n u[n] \leftrightarrow \frac{z}{z - \frac{1}{2}} + \frac{z}{z - \frac{1}{3}}$$

$$\left(\frac{1}{2}\right)^n + \left(\frac{1}{3}\right)^n$$

$$2, 0.833, \dots \leftrightarrow \frac{2z - \frac{5}{6}}{(z - \frac{1}{2})(z - \frac{1}{3})} \cdot z$$

$$\frac{z}{(z+1)(z-1)} = z \left[\frac{0.5}{z+1} + \frac{0.5}{z-1} \right]$$

partial fraction

$$-0.5 \frac{z}{z+1} + 0.5 \frac{z}{z-1}$$

$$x[n] = -0.5 (-1)^n u[n] + 0.5 \cdot 1^n \cdot u[n]$$

$$\frac{z}{z^2 - 0.8z + 0.15} \leftrightarrow z \left[\frac{A}{z-0.5} + \frac{B}{z-0.3} \right]$$

$$A = \frac{1}{0.2} \quad B = \frac{1}{-0.2}$$

$$A \cdot (0.5)^n u[n] \quad B \cdot (0.3)^n u[n]$$

partial fraction

rational function of z

$$f(z) = \frac{Q(z)}{P(z)} \quad \text{polynomial of } z$$

$$= \left(\frac{k_1}{z-p_1} + \frac{k_2}{z-p_2} + \dots + \frac{k_N}{z-p_N} \right)$$

$$p(z) = (z-p_1)(z-p_2)\dots(z-p_N)$$

$$p_i \neq p_j$$

$$p(z) = z^2 - 0.8z + 0.15$$

$$= (z-0.3)(z-0.5) \quad \text{unique roots}$$

$$p(z) = (z-0.3)(z-0.3) \quad 0.3 \text{ repeated}$$

$$\text{Ex } f(z) = \frac{z^2}{(z-0.3)(z-0.5)}$$

$$= z \frac{z}{(z-0.3)(z-0.5)} \quad \begin{matrix} (z-0.3) & z=0.3 \\ (z-0.5) & z=0.5 \end{matrix}$$

$$= z \left[\frac{k_1}{z-0.3} + \frac{k_2}{z-0.5} \right] \quad (z-0.3) \text{ cover-up}$$

$$k_1 = \frac{z}{z-0.5} \Big|_{z=0.3} = \frac{0.3}{0.3-0.5} = -\frac{3}{2}$$

$$k_2 = \frac{0.5}{0.5-0.3} = \frac{5}{2}$$

$$\text{Ex } \frac{z+1}{(z-0.5)(z-0.2)(z-0.3)}$$

$$= \frac{25}{z-0.5} + \frac{40}{z-0.2} + \frac{-65}{z-0.3} =$$

$$k_1 = \frac{0.5+1}{(0.5-0.2)(0.5-0.3)} = \frac{1.5}{0.06} = 25$$

$$k_2 = \frac{0.2+1}{(0.2-0.5)(0.2-0.3)} = \frac{1.2}{(-0.3)(-0.1)} = \frac{120}{3} = 40$$

$$k_3 = \frac{0.3+1}{(0.3-0.5)(0.3-0.2)} = \frac{1.3}{-0.2 \cdot 0.1} = -65$$

repeated roots

$$\frac{z}{(z-0.3)^2(z-0.5)} = \frac{\overset{(k_1)}{k_1}}{z-0.5} + \frac{k_2}{z-0.3} + \frac{\overset{(k_3)}{k_3}}{(z-0.3)^2}$$

$$\frac{z}{(z-0.3)^3(z-0.5)} = \frac{k_1}{z-0.5} + \frac{k_2}{z-0.3} + \frac{k_3}{(z-0.3)^2} + \frac{k_4}{(z-0.3)^3}$$

$$k_1 = \frac{0.5}{(0.5-0.3)^2} = \frac{0.5}{0.02} = 25$$

$$k_2 = \frac{0.3}{0.3-0.5} = -\frac{3}{2}$$

$$\text{let } z=0 \quad 0 = \frac{25}{-0.5} + \frac{k_2}{-0.3} + \frac{-3/2}{0.09}$$

$$k_2 = ?$$

residue (MATLAB)

$$\text{Ex } \frac{z+1}{(z-0.3)(z-0.5)(z-0.2)} = z \left[\frac{(z+1)}{z(z-0.3)(z-0.5)(z-0.2)} \right]$$

Find its inverse z-transform

$$\frac{z}{z-a} \leftrightarrow a^n u[n]$$

$$\delta[n] \leftrightarrow 1$$

$$z \left[\frac{k_1}{z} + \frac{k_2}{z-0.3} + \frac{k_3}{z-0.5} + \frac{k_4}{z-0.2} \right]$$

$$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$

$$k_1 \cdot \delta[n] + k_2(0.3)^n u[n] + k_3(0.5)^n u[n] + k_4(0.2)^n u[n]$$

$$\frac{1}{(z-1)^2} = \sum_n$$

$$\frac{z}{z-a} = \sum_{n=0}^{\infty} a^n z^{-n}$$

$$\Rightarrow \sum_{n=0}^{\infty} a^n z^{-n-1} \cdot (-n) = \sum_{n=1}^{\infty} (-n) a^n z^{-(n+1)}$$

$$\frac{z \cdot (z-a)^{-2}}{z \cdot (z-a)^{-2}} = \sum_{n=0}^{\infty} b_n z^{-n}$$

$$\left(\frac{z}{z-a}\right)' = \frac{1}{z-a} + z \cdot \frac{1}{(z-a)^2} (-1)$$

$$= \frac{(z-a) - z}{(z-a)^2} = \frac{-a}{(z-a)^2}$$

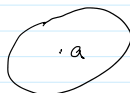
$$z^{-1}$$

$$\boxed{\frac{1}{(1-z)^2} = \sum_{n=0}^{\infty} (n+1) z^n}$$

Chapter 5 Taylor Expansion

$$f(x) = \sum_{n=0}^{\infty} a_n (x-a)^n$$

$f(z)$ is analytic



$$\oint_c \frac{f(z)}{z-a} dz = 2\pi i f(a) \quad a \in \Omega$$

$$\oint_c \frac{f(s)}{s-a} ds = 2\pi i f(a) \quad a \rightarrow z$$

$$\oint_c \frac{f(s)}{s-z} ds = 2\pi i f(z)$$

$$f(z) = \frac{1}{2\pi i} \oint \frac{f(s)}{s-z} ds$$

$$= \frac{1}{2\pi i} \oint f(s) \cdot \frac{1}{s-z} ds$$



anchor

$$\frac{1}{s-z} = \frac{1}{(s-a)-(z-a)}$$

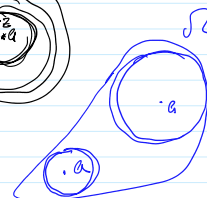
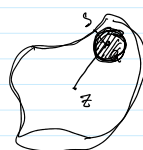
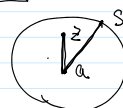
$$= \frac{1}{s-a} \sum_{n=0}^{\infty} \left(\frac{z-a}{s-a}\right)^n$$

$$\sum_{n=0}^{\infty} \left(\frac{z-a}{s-a}\right)^n = \frac{\left(\frac{z-a}{s-a}\right)^0 - \left(\frac{z-a}{s-a}\right)^{\infty}}{1 - \frac{z-a}{s-a}}$$

$$= \frac{1}{\frac{(s-a)-(z-a)}{s-a}} = \frac{s-a}{(s-a)-(z-a)}$$

$$\left|\frac{z-a}{s-a}\right| < 1 \quad \checkmark$$

$$|z-a| < |s-a| \quad \checkmark$$



$$f(z) = \frac{1}{2\pi i} \oint_c f(s) \frac{1}{s-z} ds$$

$$= \frac{1}{2\pi i} \oint_c f(s) \frac{1}{s-a} \sum_{n=0}^{\infty} \frac{(z-a)^n}{(s-a)^n} ds$$

$$= \frac{1}{2\pi i} \oint_c f(s) \sum_{n=0}^{\infty} \frac{1}{(s-a)^{n+1}} \cdot (z-a)^n ds$$

$$= \sum_{n=0}^{\infty} \left[\frac{1}{2\pi i} \oint_c f(s) \frac{1}{(s-a)^{n+1}} ds \right] (z-a)^n$$

$$= \sum_{n=0}^{\infty} c_n (z-a)^n$$

$$= \sum_{n=0}^{\infty} \underbrace{2\pi i \int_C f(s) \frac{1}{(s-a)^{n+1}} ds}_{C_n}$$

$$= \sum_{n=0}^{\infty} C_n (z-a)^n$$

$$C_n = \frac{1}{2\pi i} \oint_C f(s) \frac{1}{(s-a)^{n+1}} ds$$

$$= \frac{f^{(n)}(a)}{n!}$$

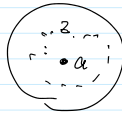


Taylor expansion

$f(z)$ is analytic, or $f(z)$ is analytic in the neighborhood of a

$$f(z) = \sum_{n=0}^{\infty} C_n (z-a)^n$$

$$C_n = \frac{f^{(n)}(a)}{n!}$$



$$f(z) = \sum_{n=0}^{\infty} C_n (z-a)^n \quad C_n = \frac{f^{(n)}(a)}{n!}$$

$$= \sum_{n=0}^{\infty} d_n (z-b)^n \quad d_n = \frac{f^{(n)}(b)}{n!}$$

$$e^z \quad a=0$$

$$(e^z)^{(n)} = e^z \Big|_{z=0} = 1$$

$$e^z = \sum_{n=0}^{\infty} \frac{1}{n!} z^n$$

$$a=1 \quad C_n = \frac{e^z}{n!} = \frac{e}{n!}$$

$$e^z = \sum_{n=0}^{\infty} \frac{e}{n!} (z-1)^n$$

$$e^3 = \sum_{n=0}^{\infty} \frac{e}{n!} z^n \Rightarrow e^2 = \sum_{n=0}^{\infty} \frac{1}{n!} z^n$$

$$e^{\sqrt{z}} = \sum_{n=0}^{\infty} \frac{e}{n!} (\sqrt{z}-1)^n$$

$$e^{\sin \theta} = \sum_{n=0}^{\infty} \frac{e}{n!} (\sin \theta - 1)^n$$

$$= \sum_{n=0}^{\infty} \frac{e}{n!} (\sin \theta)^n$$

$$e^z \quad \frac{z^2+1}{z^2+1} \text{ at } 0 \quad \sum_{n=0}^{\infty} C_n z^n$$

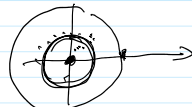
$$\frac{z^2+1}{z^2+1} \text{ at } 1 \quad \sum_{n=0}^{\infty} d_n (z-1)^n$$

$$d_n = \frac{f^{(n)}(z)}{n!} \Big|_{z=1}$$

$$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n$$

$$\text{let } z = z^{-1}$$

$$\frac{1}{1-z^{-1}} = \sum_{n=0}^{\infty} z^{-n} = \frac{z}{z-1}$$



$$-1 \cdot \frac{1}{(1-z)^2} (-1) = 1$$

$$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n \quad |z| < 1$$

$$\frac{1}{1-z^2} = \sum_{n=0}^{\infty} C_n z^n \quad z \rightarrow z^2$$

$$= \sum_{n=0}^{\infty} z^{2n} \quad C_n = \begin{cases} 1 & n \text{ even} \\ 0 & n \text{ odd} \end{cases}$$

$$1 - z^{-1} = \sum_{n=0}^{\infty} z^{-n} \quad C_n = \begin{cases} 1 & n \text{ even} \\ 0 & n \text{ odd} \end{cases}$$

$$\frac{1}{1+z} = \sum_{n=0}^{\infty} (-z)^n = \sum_{n=0}^{\infty} (-1)^n z^n$$

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}$$

$$e^z = \sum_{n=0}^{\infty} \frac{1}{n!} z^n$$

$$e^{iz} = \sum_{n=0}^{\infty} \frac{1}{n!} (iz)^n = \sum_{n=0}^{\infty} \frac{1}{n!} i^n z^n \quad (1)$$

$$e^{-iz} = \sum_{n=0}^{\infty} \frac{1}{n!} (-i)^n z^n \quad (2)$$

$$\cos z = \frac{1}{2} [e^{iz} + e^{-iz}]$$

①	1	i	$\frac{1}{2}(-1)$	$\frac{1}{3!}(-i)$	$\frac{1}{4!}$				
②	1	-i	$\frac{1}{2}(-1)$	$\frac{1}{3!}i$	$\frac{1}{4!}$				
n	0	1	2	3	4	5	6	7	8
		X		X					

$$\cos z = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} z^{2n}$$